Dear Professor Vespa,

Thank you for the opportunity to revise our manuscript titled *"Preferences for Warning Signal Quality: Experimental Evidence"* (Manuscript ID: YGAME-D-25-00181). We appreciate the thoughtful comments and suggestions provided by the AE and the reviewers which have significantly improved the paper. We have carefully addressed all points raised. Below, we provide a detailed response to each comment.

We hope these revisions meet the expectations of the reviewers and the journal. Thank you for considering our revised manuscript.

Sincerely,
Peter McGee
University of Arkansas

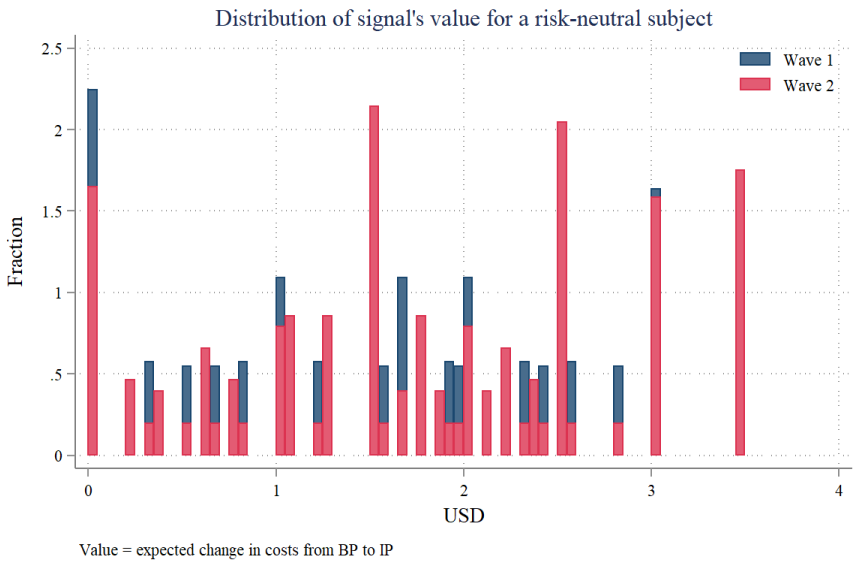
General Summary of Changes

- In response to concerns about how some signals in our original submission led to willingness-to-pay (WTPs) that are too close to the bounds, we have run additional sessions with new experimental treatments. We increased the total number of treatments for the non-Blind-Protection task from 28 to 48 treatments (Table 1) for an additional 102 subjects. Figure 1 illustrates how the addition of new treatments (in Wave 2) bolstered the set of interior WTP values for the theoretical risk-neutral agent and improved the robustness of our results.

Table 1: List of Treatments

Prior (p)	Number of Gremlins			FP rate	FN rate
	Total	Black-Swamp	White-Swamp		
	(1)	(2)	(3)	(4)	(5)
0.1, 0.2, 0.3, 0.5	2	0	0	0.00	0.00
0.1, 0.2, 0.3, 0.5	2	1	0	0.50	0.00
0.1, 0.2, 0.3, 0.5	2	0	1	0.00	0.50
0.1, 0.2, 0.3, 0.5	3	1	0	0.33	0.00
0.1, 0.2, 0.3, 0.5	3	0	1	0.00	0.33
0.1, 0.2, 0.3, 0.5	3	1	1	0.33	0.33
0.1, 0.2, 0.3, 0.5	5	1	0	0.20	0.00
0.1, 0.2, 0.3, 0.5	5	0	1	0.00	0.20
0.1, 0.2, 0.3, 0.5	5	1	1	0.20	0.20
0.1, 0.2, 0.3, 0.5	7	2	0	0.29	0.00
0.1, 0.2, 0.3, 0.5	7	0	2	0.00	0.29
0.1, 0.2, 0.3, 0.5	7	1	1	0.14	0.14

Figure 1: WTP Distribution for the Theoretical Risk-Neutral Agent



- Moreover, our empirical analysis now addresses the potential bias from the variations in the number of observations by treatment. Earlier, for example, we might have put too much weight on treatments with a truthful information structure since there will always be a total of two treatments with (one for each prior) in every session. We address the issue in two ways: (i) for the aggregate analysis, we re-weight the observations such that they are balanced by treatment; and (ii) for individual-decision analysis, we implement two-way clustering of standard errors by subject and treatment (instead of by subject only) to account for potential error correlation within treatment. Nonetheless, we find that most of our original conclusions are robust to these corrections.

- In the theory section, we removed Proposition 2 in response to a referee comment (R1,#5). We were able to sharpen Proposition 3 (currently Proposition 2) in a way that avoids approximations. The section introduction of preferences is re-written in a more rigorous way starting from expected utilities of all the strategies per referees suggestions. We also corrected a typo in the Proposition 1 proof which doesn't affect our results.

- We also revised and slightly reorganized the empirical results sections (Sections 4-6 in the original submission). First, we clarify that Section 4 provides a descriptive analysis to show that subjects reasonably understand the instructions for each task. Second, we elaborate that formal analysis to understand what drives subjects' WTP bids and the extent to which they agree with the theoretical risk-neutral benchmark is done in Section 5.

- The current Section 5.3 covers what was previously Section 6 (i.e., the discussion on mechanisms). In response to referee suggestions, we added two alternative explanations for the behavior we observed in the experiment, related to probability weighting and anticipation to the bias in the protection decision.

Response to the Associate Editor

1. The introduction of preferences is currently confusing. The expected utility (EU) formulas simultaneously present both utility calculations and optimal actions. I recommend a more incremental approach. First, derive the EU of each possible strategy (e.g., following the signal, ignoring it, doing the opposite of what signal says etc.), then identify the signal structures under which following the signal is optimal. This approach, common in information economics, may clarify your presentation and address R1's concerns about Proposition 1.

   Response: We appreciate your suggestion on the presentation of the model. In response, we have restructured the presentation of preferences starting with potential strategies and then proceeding to optimal decisions and finally value of information.

2. Can you strengthen Proposition 3 by deriving bounds for the dropped error terms?

   Response: Yes. We have now strengthened Proposition 3 by developing a proof avoiding approximations.

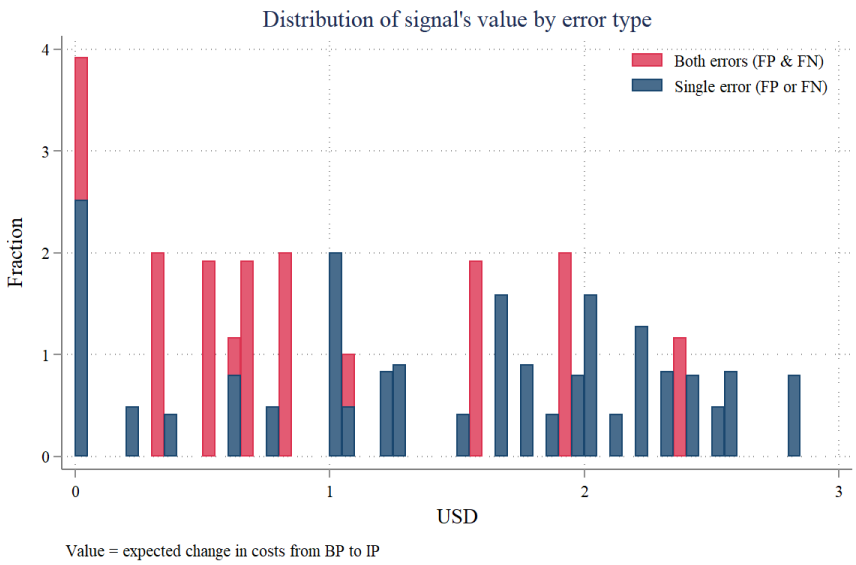
3. Reviewer 1 raises important concerns about the pooling of priors and the fact that the willingness to pay (WTP) for some signals gets too close to the bounds of the range,

which are echoed by Reviewer 2. If these concerns are indeed valid, that makes your results questionable. If feasible, consider running robustness checks by changing priors to make WTP amounts more interior or varying the elicitation method of WTP. If your lab access is limited, you may want to switch to online platforms. If you pursue this, I recommend randomizing the order of priors and incorporating Reviewer 2's comment #9 into the design as well.

Response: Thank you for these suggestions. To address the concern about having WTPs that are too close to the bounds, we have collected more data that incorporate new treatments to allow for a broader range of (theoretical) WTPs, especially for treatments with both false-positive and false-negative information structurs. We described these new treatments in our general summary of changes above. In addition, we followed the suggestion on randomizing the order of priors for the new wave of data collection.

One of the main concerns regarding the original parameter design was that the treatments with both-error conditions tend to have lower WTPs. Our new treatments address these concerns about the WTP distribution. We plot the frequency of treatments by the theoretical WTP for single-error and both-error conditions in Figure 2. It shows that our new treatments increased the theoretical WTPs for both-error conditions to include values at the higher end of the WTP distribution. We find that our original results regarding the over-/under-valuation of signals with different information structure are robust to the addition of these new treatments.

Figure 2: Theoretical WTP Distribution by Error Type



We also carefully considered R2's suggestion (#9) on reframing how gremlins are described (i.e., avoiding the term "honest"). Our formulation that uses the term "honest" follows the original framing by Coutts (2019). While the use of this term might partly explain our findings, we worry that using a new framing to collect the additional data would substantively change the experiment enough, removing our ability to credibly pool

our original and new data, resulting in the loss of statistical power. After careful deliberation, we decided to keep the original descriptions of the gremlins in the additional data collection.

4. Your individual-level analysis needs to be improved. In addition to the reviewers' suggestions in this regard, you may consider classifying participants based on the consistency of their behavior across the four parts, and test your hypotheses for each group.

   Response: Following referees suggestions, we extended individual-level analysis. First, we explored whether subjects' bias in informed protection explains the pattern of sensitivity observed in the WTP task. The data is consistent with subjects anticipating their bias, but we see no evidence that this anticipation causes the pattern of insufficient sensitivity to FP costs for low priors (FN costs for high priors) we claim as our main finding. We incorporated this analysis in Section 5.3 under the heading "Anticipation of Bias in Protection Decision" (p. 26).

   Next, we explore the consistency between reported beliefs and informed protection decisions and add the following paragraph into the paper (p. 16):

   > Subjects' choices in BE task are overall highly consistent with their choices in the IP task. We use the Mann-Whitney U-statistic to examine the number of cases when a subject inconsistently protects for a belief μ_1 but does not protect for a belief $\mu_2 \geq \mu_1$. The median U-statistic is 0.93, indicating an almost perfect rank correlation with just 1 or 2 discordant choices per subject; 35% of subjects report beliefs which are fully consistent with their protection decisions. At the same time, there is a minority of subjects with significant discrepancies: about 25% of subjects have U-statistic below 0.71 indicating multiple discrepancies.

   We also find that subjects without inconsistencies in their WTP-IP decisions exhibit higher sensitivity to FP and FN costs indicating more attention to signal characteristics, though these differences are not always statistically significant (the tables are in the response to comment 10 by R1). Consistent subjects still exhibit low sensitivity to FP costs for low priors and high sensitivity to FN costs for high priors, which aligns with our previously reported findings. We decided against including this discussion into the paper, given that the hypothesis lacks clear theoretical grounds and our findings are not affected.

5. Tables 2 and 3 are difficult to interpret. Please consider a more transparent way of presenting the data across treatments. The reviewers offer constructive suggestions here.

   Response: Thank you for this valuable suggestion. We have now clarified that the purpose of Section 4 (and therefore, the associated Tables 2 and 3) is to use descriptive analysis to verify that subjects have a reasonable understanding of our instructions for each task. We have therefore added the following paragraph in the beginning of Section 4:

   > Given the complexities of some of these tasks, this section illustrates the extent of subject comprehension using simple descriptive analyses. We find that subjects understand these tasks reasonably well. This section also provides a

qualitative overview of how subjects tend to deviate from the theoretical predictions for each task. We provide a more formal regression analysis of how signal characteristics affect subjects' WTP later in Section 5.

At the same time, we incorporated the suggestion of R1 (#7) to improve these tables by disaggregating subject decisions further by prior. These tables are currently in the appendix. Upon inspection, we feel that further disaggregation by prior makes the tables harder to read and does not improve the clarity of the exposition. However, we would be happy to accommodate the editor suggestion by moving these table to the main text as needed.

6. I agree with Reviewer 1 regarding the terminology. Consider using "information source" or "information structure" for each treatment, and reserve "signal" for the realized signal.

   Response: Thank you for this helpful suggestion to improve the clarity of our exposition. We have now revised our text as suggested using "signal" to refer to the realized signal, and "information structure" to refer to the signal's characteristics.

7. Introduce the definition of imprudent in the main text rather than only in the proof.

   Response: Thank you for this suggestion. One of the referee comments (R1, #5) prompted us to re-evaluate potential applications of Proposition 2 and eventually to omit it from the paper. Indeed, the assumptions of the original Proposition 2 (i.e., risk averse and imprudent) are rarely satisfied and the rest of the paper does not require this proposition.

8. Please cite the papers suggested by the reviewers.

   Response: We have added these references to the relevant discussions in Section 5.3 (text on p. 25 and footnote 20 on p. 27).

Reviewer 1:

We would like to thank you for your comments and suggestions. We appreciate the time and care you took to write them. Thanks to these comments, we have made revisions that we believe have significantly improved the paper. Below is our detailed response to your comments:

1. First, the term "signal" used by the authors is somewhat confusing. When it is ex-post, the signal is the realization from an information source, while when it is ex-ante, it is the source itself. Given that the experimental design contains elicitations regarding both expost signals and ex-ante sources, I would like to see a clearer exposition of this terminology. In general in the literature, signal or news are often referred to as signal realizations, and ex-ante it is called the information source. I will follow this custom in this report.

   Response: Thank you for this helpful suggestion to improve the clarity of our exposition. We have now revised our text as suggested using "signal" to refer to the realized signal, and "information structure" to refer to the signal's characteristics.

2. On page 6, the assumption that "access to a signal can increase expected utility if the signal affects her protection decisions" influences the calculation of the WTP b in equation (1), as well as the partial derivatives for sensitivity. I do not think this assumption is proper here. The assumption should be more fundamental and made on the properties of the information source (P_{ij}), not on behaviors. In fact, in the set of parameters chosen for the experiment, four tasks did not satisfy this assumption (non-instrumental information sources). Another suggestion is to verify whether subjects' choices indeed follow this assumption. It provides a measurement at the individual level for contingent reasoning. If a significant number of subjects put a positive WTP for the non-instrumental information sources, then the theoretical analyses based on this assumption become questionable.

   Response: We appreciate the nudge to look again on the exposition of the model. This sentence was never meant to be read as an assumption, hence we decided to rewrite the introduction of preferences in a more gradual way, starting from utilities from different strategies and then showing under what conditions having a signal improves expected utility for expected utility-maximizing subject.

   From the theoretical point of view, there are no informative information structures which are non-instrumental for all the EU-maximizing decision-makers. Under our assumption of $P_{11} > P_{01}$ the posterior probability of an adverse state is higher when the signal is black: $\mu(S = B) > \mu(S = W)$. Subjects protect when $u(Y-c) > \mu(S)u(Y-L)+(1-\mu(S))u(Y)$ or $\mu \geq \frac{u(Y)-u(Y-c)}{u(Y)-u(Y-L)}$. We can fix $u(Y-L) = 0$, $u(Y) = 1$ and then find $u(Y-c)$ such that $0 \leq 1-\mu(S = B) < u(Y-c) < 1-\mu(S = W) \leq 1$, so the subject will protect after receiving a black signal, but wouldn't protect after receiving a white signal.

   From the empirical point of view, we indeed find that a significant number of subjects report positive WTP for information sources they do not use. We elaborate this finding in our response to your concerns about data analysis (comment #10 below) and reported this result in Section 5.3 in our discussion whether preference for non-instrumental information explains our empirical results (p. 25).

3. Also in Appendix B.1 Proposition 1, the last equation is not correct. It should be $\leq$ not $=$. The statement "Because expected utility with a signal is a decreasing function of b_0 we obtain $b > b_0$" should be $b < b_0$."

   Response: Thank you for pointing this out! We corrected the typo at the end of the proof making it consistent with the Proposition. We also added a strict aversion assumption which together with payoffs non-degeneracy (true as long as $\pi > 0$) justify strict inequality. The proposition itself is correct and now matches the proof.

4. On page 7, I do not understand the prediction that "the model of a risk-neutral agent suggests that subjects' WTP should have equal sensitivity to costs from false-positive and false-negative signals (equation 2)." In equations (2)-(4), the partial derivatives of the WTP b with respect to FP and FN are different and only equal when the prior is 0.25. But the experiment tasks have varying priors from 0.1-0.5. As a result, Hypothesis 2 is not correct and the analyses for Result 2 should be restructured.

   Response: We appreciate your comment which indicates to us that the distinction between false-positive/false-negative rates and costs was not clearly outlined in the paper. While FP (P_{10} in the model) and FN rates (P_{01}) depend purely on the information structure, FP and FN costs account for expected protection costs and expected losses due to lack of protection. We now added footnote 1 and a paragraph on p. 6 to define FP and FN costs clearly: *"Note also that WTP is always equally affected by FP and FN costs which are defined as $\pi(1 - P_{01})c$ and $\pi P_{01}L$, respectively."*

5. Propositions 2 and 3 rely on quite restrictive assumptions and simplifications. I would like to see discussion of the generalizability of these propositions, perhaps with simulations or numeric methods.

   For instance, Proposition 2 relies on the "imprudence" assumption that might be inconsistent with standard utility specifications and the authors' own empirical findings. The condition $U'''(w) < 0$ fails to hold for the most commonly used utility functions in economics: CRRA utility has $U'''(w) > 0$ for reasonable risk aversion parameters $\gamma > 0$, while CARA utility has $U'''(w) > 0$ for any $\alpha > 0$. In the analyses for BP, the authors acknowledge that many of our subjects (24) start protecting at lower priors (0.05-0.15), indicating strict risk aversion. A smaller group of subjects makes choices consistent with risk loving by protecting at a probability of 0.3." There should be further discussion of the validity of the imprudence condition.

   Response: Thank you for pointing this out. In revising the paper to respond to this comment, we realized that no subject with CRRA utility function can meet the condition for this proposition. Given lack of applicability, we removed Proposition 2 from the paper. The rest of the paper is unaffected since we never use it to inform our empirical analysis.

6. Proposition 3's derivation relies on a first-order Taylor approximation that drops higher-order terms without establishing error bounds or validity conditions. The proof provides no justification for why the neglected terms $o(b)$ and $o(b + c)$ are "small enough" relative

to the retained terms. For meaningful values of b and c (5protectioncost; upto5 WTP), the neglected terms could be substantial. I would like to see error bounds demonstrating when the approximation is valid.

Response: Thank you for your suggestion. We have rewritten the proof in a more rigorous way that avoids any approximations.

7. I hope the authors can revise Tables 2 and 3 accordingly since the pooling of priors and FN and FP structures may be uninformative. Given that, it will be more straightforward to check how the elicited posteriors, protection actions and WTPs change for different priors and error types.

   Response: Thank you for this valuable suggestion. We have now clarified that the purpose of Section 4 (and therefore, the associated Tables 2 and 3) is to use descriptive analysis to verify that subjects have a reasonable understanding of our instructions for each task. We have therefore added the following paragraph in the beginning of Section 4:

   > Given the complexities of some of these tasks, this section illustrates the extent of subject comprehension using simple descriptive analyses. We find that subjects understand these tasks reasonably well. This section also provides a qualitative overview of how subjects tend to deviate from the theoretical predictions for each task. We provide a more formal regression analysis of how signal characteristics affect subjects' WTP later in Section 5.

   At the same time, we incorporated your comment to disaggregate subject decisions further by prior. These tables are currently in the appendix. Upon inspection, we feel that further disaggregation by prior makes the tables harder to read and does not help clarify the exposition. However, we would be happy to accommodate the suggestion to move these table to the main text as needed.

Experimental Design Concerns

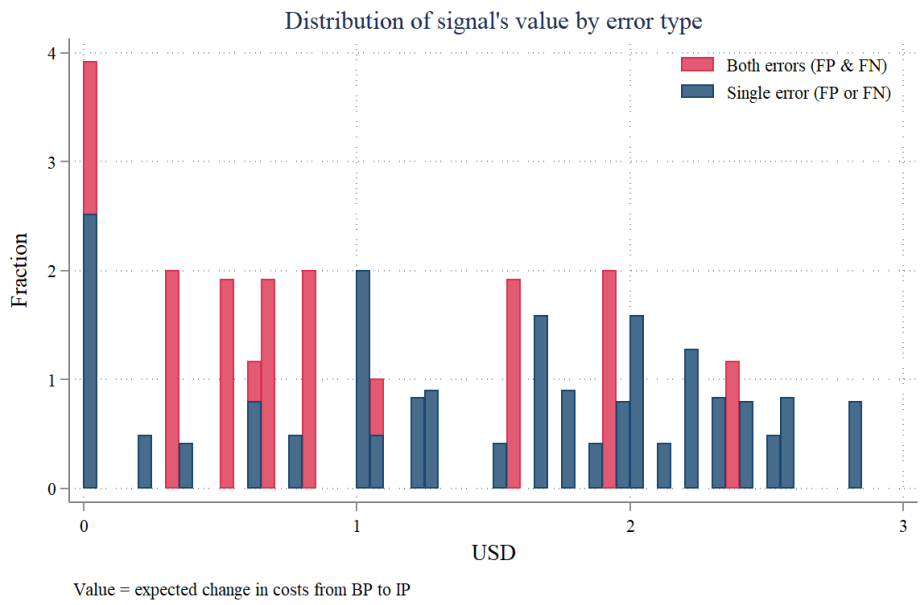
8. My main concern about the experiment is the parameter designs. As we observe from my calculations, many conditions produce extreme values. I am worried that some key findings appear to be driven by systematic measurement artifacts created by the experimental parameter choices rather than genuine behavioral patterns. For instance, on page 16, Result 1 shows that both-error conditions have systematically lowest WTP. This pattern might be suspicious since single-error conditions often produce extreme posteriors (0 or 1) while both-error conditions tend to produce intermediate posteriors. The complexity level is different. Likelihood insensitivity, rather than belief updating, might also explain the valuation. In addition, almost all the both-error conditions generate very low WTPs, thus the apparent overvaluation for them might "simply be due to reversion to the mean."

   Response: Thank you for this suggestion. We address this concern by expanding our treatments from 28 to 48 treatments, to allow for a larger set of interior (theoretical) WTP. The new treatments also include more information structure with both (false-positive and

false-negative)-error conditions.

Our new treatments address these concerns about the WTP distribution. We plot the frequency of treatments by the theoretical WTP for single-error and both-error conditions in Figure 4. It shows that our new treatments increased the theoretical WTPs for both-error conditions to include values at the higher end of the WTP distribution.

Figure 3: Theoretical WTP Distribution by Error Type



In addition, we address the potential imbalance in the treatment frequency from our treatment assignment protocol. Since not all treatments have the same frequency, we may be concerned that our results are driven by more frequent treatments. For example, since there are always a total of two truthful treatments with (one for each prior) in every session, we might have put too much weight on treatments with a truthful information structure. We address the issue in two ways: (i) for the aggregate analysis, we re-weight the observations such that they are balanced by treatment; and (ii) for individual-decision analysis, we implement two-way clustering of standard errors by subject and treatment (instead of by subject only) to account for potential error correlation within treatment.

We find that our original results regarding the over-/under-valuation of signals with different information structure are robust to the addition of these new treatments as well as our revised (and more conservative) empirical strategy that accounts for the treatment assignment process.

9. This may also drive the key conclusion about asymmetric sensitivity. For low prior conditions, the low base rate ($\pi = 0.1$) means theoretical WTP is already near zero for most signals. Adding false positive costs should theoretically drive WTP even lower toward zero, but there is a boundary effect where subjects cannot bid negative amounts, so any positive bid appears as insufficient reduction." This creates a measurement artifact where the failure to reduce WTP enough" is simply because observed WTP is bounded below

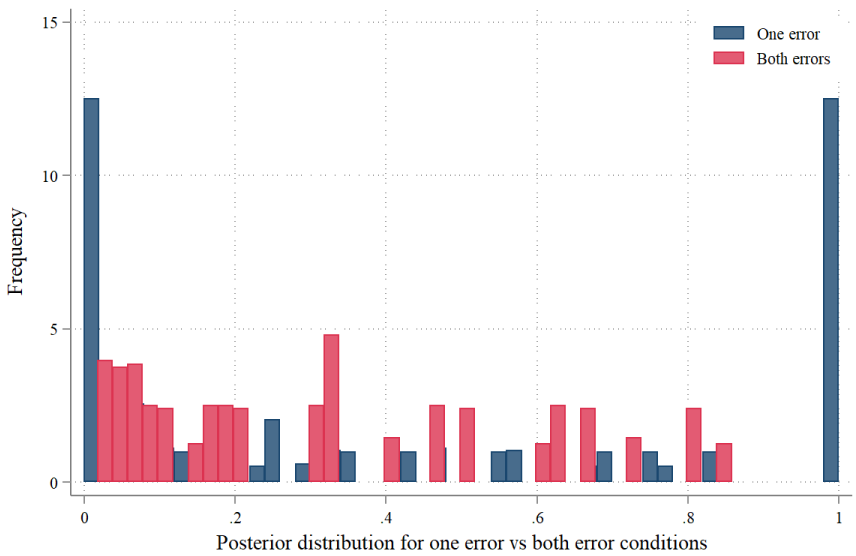
at zero. For high prior conditions, the high base rate means theoretical WTP could be substantial ($3-5), and adding false negative costs should theoretically reduce WTP from these high levels. However, subjects may be reluctant to bid close to their $30 endowment or have budget constraints, creating a measurement artifact where the "underreaction" is because observed WTP is practically bounded above.

Response: We share the concern about the boundary effects. To examine the robustness of our results to the boundary effects, we estimated a tobit regression of the discrepancy (accounting for these boundaries) on FP and FN costs. Appendix Table A7 presents the results. We show that our results from the linear regression (from Table 5) were robust to these potential boundary effects. We also show that most of our treatments correspond to interior values of information for risk-neutral subjects, particularly after extending our sample in the second wave.

The analyses and conclusions of this paper might be stronger if the FN and FP combinations also induced some intermediate posteriors. Otherwise, I would suggest the authors have a discussion about how the parameter design may create artifactual results.

Response: To address this concern, we used a new wave of data collection to increase the proportion of treatment with intermediate posteriors. Currently, the majority of treatments has intermediate posteriors. Given that we still have many "honest" treatments used to introduce subjects to changing priors, we report our results with clustering also on treatment level leading to more conservative standard errors.

Figure 4: Theoretical WTP Distribution by Error Type



Data Analysis Issues

10. The data analyses are not well-structured for readers to grasp the takeaways. Most results use aggregate statistics that mask individual inconsistencies. They often pool subjects across tasks rather than tracking individual patterns and focus on population averages

rather than individual coherence.

Response: Thank you for this comment, which to our understanding, is related to comment 7 above. We apologize that we did not clarify in our original submission that Tables 2 and 3 (and the associated Section 4) are intended to show — using descriptive statistics — that on aggregate, subjects make sensible choices in novel tasks. We also would like to illustrate population-wide facts on preferences over warning signals which could be useful to designers of these systems who often have no access to individual-level data on risk-preferences and probability updating.

We defer the more formal, individual-decision-level regression analysis to Section 5, which includes examining how responses vary with risk preferences, belief accuracy, and demographic variables. As we described in comment #7 above, we hope the paragraph at the beginning of Section 4 helps to clarify the intent of the aggregate analyses and address your concern.

A missing piece of the data analyses is cross-task consistency checks. The authors' experimental design contains four parts, yet they analyze each component in isolation without examining individual-level consistency across tasks. This represents a significant missed opportunity for validation, as their theoretical framework explicitly requires subjects to integrate these components coherently.

- For instance, a BE-IP consistency check could examine whether, given a threshold posterior of 0.25, a consistent decision maker who reports posterior > 0.25 in the BE task also chooses "Protect" in the IP task. Across the six tasks generating 12 posteriors, the authors could identify decision thresholds for different subjects and inconsistency measures at the individual level.
- Similarly, an IP-WTP consistency check could examine whether participants who chose to protect or not protect regardless of possible signals assign zero WTP to information structures, as predicted by the assumption that access to a signal can increase expected utility if the signal affects her protection decisions." These analyses would likely reveal that high-consistency subjects show clearer false positive/false negative sensitivity patterns, while low-consistency subjects drive the null aggregate results, explaining why their current pooled analysis fails to detect the theoretically predicted effects.

Response: Following your suggestions, we extended our analysis with additional cross-consistency checks. First, we explore the consistency between reported beliefs and informed protection decisions and add the following paragraph into the paper (p. 16):

> Subjects' choices in BE task are overall highly consistent with their choices in the IP task. We analyze this consistency by using Mann-Whitney U-statistic which in our case maps to the number of cases in which a subject inconsistently protects for belief μ_1 but doesn't protect for belief $\mu_2 \geq \mu_1$. The median U-statistic is 0.925 indicating almost perfect rank correlation with just 1 or 2 discordant choices per subject, and 35% of subjects report beliefs which are fully consistent with their protection decisions. At the same time, there is a minority of subjects with significant discrepancies - about 25% of subjects have U-statistic below 0.71 indicating multiple discrepancies"

Next, in response to R2, we study whether the subjects' bias in the informed protection task explains the pattern of sensitivity to FP and FN costs we observe in the WTP task (new subsection "Anticipation of Bias in Protection Decisions" on p. 26). We create two new variables to measure discrepancies between optimal and actual protection decisions and find out that while these discrepancies correlate with reported WTP, taking them into account does not explain the pattern of sensitivity to FP and FN costs we observe.

Answering your comment on the use of non-instrumental information, the majority of decisions are consistent across tasks, though in many cases subjects do pay for non-instrumental information.

	WTP = 0		WTP > 0	
	Freq	Mean WTP	Freq	Mean WTP
React to a signal	128	0	706	2.32
Don't react to a signal	101	0	295	1.97

Based on decisions in WTP and IP tasks, we create a leave-one-out definition of a consistent subject, then conduct heterogeneity analysis based on this subject consistency measure. Subjects without inconsistencies in their WTP-IP decisions exhibit higher sensitivity to FP and FN costs indicating more attention to signal characteristics, though these differences are not always statistically significant (see the table below). However, even consistent subjects exhibit low sensitivity to FP costs for low priors and high sensitivity to FN costs for high priors, which aligns with our previously reported findings.

Table 2: Deviations from Signal Value (WTP - Value) and Protection Consistency

	All		Prior	
			$\{.1, .2\}$	$\{.3, .5\}$
FP costs	0.421	0.541	0.511	0.427
	(0.258)	(0.274)*	(0.103)***	(0.113)***
FN costs	0.287	0.327	0.123	0.424
	(0.117)**	(0.132)**	(0.111)	(0.052)***
Consistent WTP=1 × FP costs		-0.250	-0.094	-0.425
		(0.100)**	(0.117)	(0.160)**
Consistent WTP=1 × FN costs		-0.074	-0.188	-0.128
		(0.089)	(0.124)	(0.059)**
R^2	0.480	0.483	0.720	0.731
Obs	1230	1230	615	615
Consistent Subjects:				
False Positive		0.291	0.416	0.001
se		(0.252)	(0.118)	(0.147)
p-value		[0.254]	[0.002]	[0.993]
False Negative		0.253	-0.065	0.296
se		(0.121)	(0.136)	(0.042)
p-value		[0.042]	[0.634]	

11. In the key section "WTP and Signal Characteristics," the authors use the difference between individual WTP and risk-neutral WTP for regressions. With the decisions in other tasks, I think it makes more sense to use the difference between observed WTP and predicted WTP, which should be a function of individual risk attitude from BP, individual beliefs from BE, and individual action plans from IP. The function could be a simple OLS but could also incorporate behavioral elements that account for different biases in each component of the tasks.

    Response: Thank you for your suggestion. Indeed, we considered implementing this approach — to wit, constructing predicted WTP as a function of individual risk attitudes (BP), beliefs (BE), and action plans (IP) — for our original submission. We decided against implementing this approach for three reasons. First, constructing a benchmark WTP that accounts for risk preferences would require choosing a functional form and parameter values, making our benchmark model-dependent. Second, this approach will introduce more noise to the benchmark value since parameter estimates would rely on just 6 blind protection choices that are not always internally consistent. Finally, we believe a risk-neutral benchmark is more applicable to practical settings, where individual risk-preference parameters are typically not observed and benchmark valuations often assume risk neutrality.

12. For section 6, the evidence provided for the key conclusion is not convincing enough. The claim that people cannot distinguish false positive from false negative rates relies mainly on failing to find a significant difference in how people respond to FP versus FN costs, but failing to find a difference does not prove that no difference exists. It could simply mean their test lacks sufficient power or that their data is too noisy. More importantly, their approach contradicts their own theory since they average responses across all subjects and conditions, but their theory predicts that different types of people (risk-averse versus risk-neutral) should show different patterns of FP/FN sensitivity. By pooling everyone together, they make it impossible to detect the very differences their theory expects to find. The authors also never directly test whether people actually use a "total error rate" strategy but simply assume this explains their results without considering other possibilities like measurement problems or task confusion.

    Response: We understand your concern that failing to reject a difference between responsiveness to FP and FN rates does not, by itself, establish that no difference exists. We agree that "null results" must be interpreted carefully. In the following, we describe how we have responded to each point of concerns above.

    Statistical power. To address this issue, we collected a second wave data collection which almost doubled our sample size increasing it from 105 to 207 subjects. Increasing the sample tightened confidence intervals for our estimates, but we still fail to find significant differences in responsiveness to FP and FN rates not for one but for four different priors used in our experiment (see Figure 5). Our main results as reported in Table 5 also remain unchanged with lower overall sensitivity to FP and FN costs and pattern of higher sensitivity to FN costs for low priors.

(Non-statistical) noise. Section 4 shows that most subjects understood the instructions and were not confused. Informed protection decisions are monotonic in posterior probabilities and generally consistent with protection patterns in both blind protection decisions and elicited beliefs in belief elicitation. Belief errors are mostly minor though we do have several subjects reporting opposite beliefs for certain signals which is consistent with task confusion.

Subject heterogeneity. We address the concern over subject pooling and heterogeneity directly in several ways. Table 5 in our paper and Appendix Tables A6 and A7 in the Appendix report how responsiveness to FP and FN costs changes with risk preferences and beliefs accuracy. We report heterogeneity with respect to risk preferences that is consistent with our theory. This result emerges due to the increased statistical power from the additional data collection. However, the sensitivities to FP and FN costs change with priors even when accounting for risk preferences. We also study heterogeneity with respect to gender, education, and taking statistical classes, and find no significant heterogeneity in responsiveness though some effects on levels (see Appendix Table A5).

Evidence for the inability to discriminate FP/FN. Our belief that many subjects confuse false-positive and false-negative costs also comes from tests conducted for belief elicitation and informed protection tasks. In belief elicitation task we see that false-positive rates significantly increase belief error when the signal is white, while false-negative rates do the same with black signals. This pattern is best explained with partial confusion between using false-positive and false-negative rates. Next, in informed protection subjects start to protect when false-positive rates become positive even when the signal is black, which confirms the pattern. Finally, many subjects explicitly state using the proportion of black-eyed and white-eyed gremlins as their main decision-making heuristic.

Contribution and Literature

13. The contribution is not clearly pointed out within the field of experimental studies for the demand for information sources. For instance, the failure of distinguishing the value of FP and FN is also implied in Xu (2022) while also accounting for the effects of risk attitudes and belief updating biases. I want to see a clearer comparison and innovation of this paper. Some other key references are missing, for instance, Masatlioglu, Orhun, and Raymond (2023), Charness, Oprea, and Yuksel (2021), Montanari and Nunnari (2023), which all study skewed information sources, and some of their skewness conditions have the same structure as the FP and FN rates studied in this paper.

Response: Thank you for this comment which helped to clarify the contribution and to incorporate additional related work. Our main contribution is in studying demand for information involving protection decision which is different from previous literature on the value of information. We have rewritten the discussion of our contribution in the following way (emphasis added):

> Second, we contribute to the emerging experimental literature on the demand for information by studying signals that affect protection decisions. Previous

> studies in this literature employ prediction games, in which subjects must guess an optimal state under uncertainty (Hoffman, 2016; Ambuehl and Li, 2018; Xu, 2022; Montanari and Nunnari, 2023). Generally, they find that while the demand for information increases with signal quality, it increases more modestly than expected from a Bayesian decision maker. Two of these studies employ laboratory experiments. Ambuehl and Li (2018) find that subjects underreact to the accuracy of a binary signal about the state of the world, but put a premium on completely certain signals. Xu (2022) shows that many subjects choose non-instrumental signals over instrumental signals, consistent with failures in contingent reasoning about the future value of information and with partial failures to distinguish FP and FN rates. *Both studies employ a prediction game.*
>
> We extend this literature by moving beyond a prediction game, and instead incorporating protection decisions with three distinct payoffs (i.e., full payoff, full payoff minus protection costs, and full payoff minus losses). With asymmetric payoffs, risk preferences affect the value of information and can change sensitivities to false-positive and false-negative rates. Asymmetric payoffs also creates asymmetry in how subjects value different types of errors which is not observed in (symmetric) prediction games.

We now separately cite some papers on preferences for non-instrumental information/anticipation bias on p. 25:

> For example, Eliaz and Schotter (2010) find that subjects are willing to pay to know the probability of their choice being correct even if this information cannot affect their choice, while Ganguly and Tasoff (2017) document that most people are willing to pay a small amount to know their pre-determined experimental payoffs at the beginning of the experiment rather than at the end. Similarly, Masatlioglu, Orhun and Raymond (2023) reveal a strong and systematic preference for positively skewed information, even when such signals are less informative, suggesting that anticipatory emotions such as hope and anxiety shape information demand independently of instrumental value.[1]"

Another strand of the literature missing is ROC (Receiver Operating Characteristic) analysis, which is a statistical method used to evaluate the performance of binary classification models, particularly in radiology, diagnostic testing and machine learning. The structure is relevant. I hope there would be more exploration of how the results in this paper relate to findings in this literature. Please see Fawcett (2006) for a review.

Response: Thank you for this suggestion. We are aware of ROC literature, which is highly relevant in terms of terminology used, but it is primarily concerned with evaluative statistics and methodology. We do not feel that we can frame our contribution as relevant for this literature, but included Fawcett (2006) citation to give context for the key tradeoffs in our paper and to allow interested readers to explore further.

[1] Unlike Masatlioglu, Orhun and Raymond (2023), Oliveros, Zultan and Llorente-Saguer (2025) find that decision makers exhibit a striking and persistent preference for symmetric sources, i.e., those offering a balanced distribution of positive and negative signals, even when asymmetric sources would yield more decision-relevant information. The authors argue that symmetry exerts an intrinsic appeal, suggesting that individuals prefer information structures that "feel fair" or representative, independent of their actual informativeness.

Minor Comments.

14. Please indicate significance of the estimated parameters in your regression tables for better reading experience.

    Response: Thank you for pointing this out. We apologize for the absence of the statistical significance stars, which we forgot to put back after being removed due to an editorial requirement of another journal. We have now put the significance stars in all regression tables.

15. On page 1, the sentence "While the optimal threshold depends on user preference over the costs of these probabilistic errors, currently there is no guidance on what this threshold might be beyond assuming that decision-makers weigh false-positive and false-negative costs equally" is not consistent and not true. The optimal threshold depends on the costs of the false-positive and false-negative probabilistic errors, which in most cases are not weighted equally, such as in medical tests where the consequence of missed detection is weighted more than the consequence of false alarm.

    Response: In the original version of the paper, we didn't clearly explain the difference between false-positive (FP)/ false-negative (FN) rates and FP/FN costs. The costs in our terminology already account all the costs scaled to probabilities of outcomes and so designed to be weighted equally by any risk-neutral agent. We added the following clarification in the theory section: *Note also that WTP is always equally affected by FP and FN costs which are defined as* $\pi(1 - P_{01})c$ *and* $\pi P_{01}L$*, respectively.*

16. On page 8, the IP task description is not clear. It states "subjects must make a protection decision given the prior probability of drawing a black ball. Subjects learn a prior and signal's accuracy," but subjects should make two decisions about protection, each for a possible hint.

    Response: Thank you for this really helpful suggestion. We have revised our description of the task as follows, with the clarification that subjects must make two protection decisions (at the end of the paragraph, emphasis added)

    > As in the BP task, subjects must make a protection decision before learning if they drew a black ball. Prior to the protection choice, subjects learn the prior probability of drawing a black ball, receive a signal about the ball, and learn the signal's information structure... *Subjects are asked to make two protection decisions, to wit, for black and white signals.*

17. On page 9, there is a concerning inconsistency between the belief elicitation mechanism described in the paper and the actual experimental instructions given to subjects. The paper claims to follow "the stochastic version of the Becker-DeGroot-Marshak mechanism developed by Grether (1992) and Holt and Smith (2009)" with a specific mathematical formulation involving uniform random draws and conditional losses. However, the experimental instructions describe a completely different mechanism that simply tells subjects "you make more money if your guess is closer to the actual probability of the

event" with examples using different probability values and payoff calculations that bear no resemblance to the theoretical mechanism described in the paper. This discrepancy raises serious concerns about the validity of the belief elicitation data. Moreover, the instructions themselves appear potentially confusing to subjects, particularly the statement about "the actual probability of the ball being black is 10 percent" when each individual ball is deterministically either black or white. If the goal is incentive compatible belief elicitation where accuracy is rewarded, it would be much clearer to implement a proper scoring rule that directly penalizes prediction errors rather than using this convoluted mechanism that appears to differ between the paper's description and the actual implementation.

Response: We considered proper scoring rules when designing the experiment but decided against them because they require subjects to be risk-neutral (Holt and Smith, 2009). Instead, we use the stochastic BDM as described in Karni (2009). This method does not require risk neutrality to elicit truthful responses, but it is more difficult to explain. Moreover, our instructions incorporate the lessons from Danz, Vesterlund and Wilson (2020) who find that just explaining that reporting honestly maximizes the payoff results in more accurate reported beliefs than explaining the payoff algorithm.

Hence our instructions start with the claim of honest reports maximizing expected payoffs, gives a simplified example, and only then proceed to the actual mechanism:

> If any round of this task is chosen as the Payment Round, the computer would, first, draw a ball at random from the Box. Then it would record a hint from one randomly chosen gremlin from the group. Finally, it will draw one random lottery with chances between 0 and 100.
>
> This computer will then calculate your payment based both on the hint, the actual ball color and this random lottery. This is easier to understand through an example. Suppose, that the gremlin hints that the Ball is white and you estimate that the Ball is indeed white with probability 85%. If a computer draws a lottery with chances of 85% and above, then you lose \$20 if the Ball is white. If the computer draws a lottery with chances lower than 85%, then you would lose \$20 with the chance specified in the lottery.

Our example follows the algorithm description as it is provided, for example, in Karni (2009):

> A bet on an event E is a simple act that pays x dollars if E obtains and y dollars otherwise, $x > y$, and is denoted by $x_E y$... The elicitation mechanism selects a random number r from a uniform distribution on $[0, 1]$ and requires the agent to submit a report, $\mu \in [0, 1]$, of his subjective probability assessment of the event E. The mechanism awards the agent the payoff $\beta := x_E y$ if $\mu \geq r$, and the lottery $l(r, x, y)$ if $\mu < r$.

Our example maps to this description, except for the inversion of payoffs to losses: $\mu = 0.85$, $r =$ the chance with which a computer draws a lottery. If $\mu \leq r$, subject is losing L if the ball is white (event); otherwise, subject plays a lottery with a chance r of losing L. The inequality is flipped to adjust to the loss domain. This lottery example can be

reformulated equivalently in terms of gains which aligns with the theoretical description more precisely. Losing only on white is equivalent to the subject winning \$20 if the ball is black. Belief of the ball being black is $\mu' = 1 - \mu$. Then let $r' = 1 - r$. The payoff is determined by the ball being black if $\mu \leq r \implies 1 - \mu' \leq 1 - r' \implies \mu' \geq r'$. Otherwise the subject wins \$20 with probability r'.

The statement that "the actual probability of the ball being black is 10 percent" appears in the first introductory example designed to build intuition. It refers to the objective probability of drawing a black ball from a box with a known composition, not to the uncertainty about the realized ball once drawn. We agree that taken in isolation this phrasing could appear imprecise; however, it is immediately followed by a more detailed and exact explanation of the stochastic mechanism, which we reproduced above and which corresponds directly to the formulation in Karni (2009).

18. Given the central finding on asymmetric sensitivity depending on the prior probability, I think the comparative statics of the sensitivity for different priors is missing in this paper. It is easy for the risk-neutral benchmark but is not straightforward with risk attitudes.

    Response: Thank you for pointing this out. Indeed, this would have been an interesting result, but as you pointed out it is not straightforward when utility slopes can potentially change with bids and bids change with priors.

19. On page 14, "We define updating errors as the difference between the subjects' elicited belief and the actual posterior probability of drawing a black ball for a given signal" will be understood by most people as Updating Errors = Elicited Belief - Actual Posterior Probability, which is not consistent with your figures and follow-up descriptions.

    Response: Thank you for this correction. This was an error on our part. We have corrected all references in the text, tables and figures such that updating errors are defined as $error = belief - posterior$ throughout.

20. There are also some ambiguous pronoun references throughout the paper. On page 15, This suggests that subjects understand the signal structure but have difficulty implementing the optimal strategy" leaves unclear what "This" refers to, since the preceding sentence discusses multiple results.

21. Similarly, on page 17, These results are consistent with our hypothesis but contradict previous findings" does not specify which results or which previous findings. Page 18 references low WTP conditions" and "high WTP conditions" without clearly defining the threshold or providing descriptive statistics.

    Response: We believe these last two minor comments (#20 and #21) might have been mistakenly included here, as we cannot find the sentences they refer to in our original submission.

Reviewer 2:

We would like to thank you for your comments and suggestions. We appreciate the time and care that you have spent in writing them. Thanks to these comments, we have made revisions that we believe have significantly improved the paper. Below is our detailed response to your comments:

1. Empirical analysis of how future protection actions affect WTP:

   a) I believe that, compared to existing studies on the demand for information, the most novel element of this paper is the inclusion of protection actions. As the authors mentioned, the existing literature employs prediction games; thus, states are usually symmetric, and guessing the underlying state is relatively trivial. Here, the protection action is asymmetric, and protection decisions become non-trivial, as we have known in the literature on choice under uncertainty. As future protection actions affect individuals' WTP for information, the presence of protection action would have a non-trivial effect on subjects' WTP.

   b) However, the current version of the paper does not explore much of the relationship between protection actions and WTP. Therefore, I encourage the authors to investigate how the protection actions observed in the experiment affect WTP, which sets the paper apart from the existing literature. The analysis would also lead to new implications. Intuitively, WTP for signals is roughly affected by two factors: 1) the understanding/preference (etc.) over information 2) anticipated protection action taken by the subject's future self. As shown in Table 7, subjects exhibit failure to distinguish FP and FN even when choosing their protection actions. Is the equal sensitivity of WTP w.r.t. FP and FN driven by rational anticipation of the "bias" in protective choice? Or is it driven by the heuristic when subjects compute the expected benefit of getting the signal? Depending on the answer, the result will have different policy implications for encouraging the acquisition of warning signals.

   Response: Following your suggestion, we added a subsection to study the connection between subject's bias in the informed protection as potential explanation for pattern of sensitivities to FP and FN costs in the WTP task (p. 26). We find that actual costs elicited by subjects have some predictive power for willingness-to-pay decisions, but fall short of explaining our main results. Overall, we observe that many subjects approach tasks more independently resulting in low though significant correlations across the tasks.

2. Some re-ordering of the results, and exploring Figure 5:

   a) Result 2 on page 18 is interesting. However, it is a comparison of subjects' WTP with the risk-neutral benchmark, and it is unclear what implications we can draw from it. One could also question whether risk neutrality is a meaningful benchmark. I understand the novelty and importance of the results only until the end of the paper, where the authors discuss the subjects' failure to distinguish FP and FN rates. Figure 5 on page 22 is particularly clear and striking. I suggest that the authors move the discussion of subjects' failure to distinguish FP and FN rates to right after result 2, and frame it as an analysis on the underlying mechanism. The rest of the discussion could be framed as a section discussing/ruling out alternative explanations.

Response: We thank the referee for this very valuable suggestion. This suggestion has helped us rethink the organization of the paper to link Result 2 more clearly with the rest of Section 5.2. We have now organized the result such that Section 5.2 focuses on describing how the the deviations from the theoretical — risk-neutral, Bayesian updating — agent are correlated with FP and FN costs, and explore whether risk preference and updating errors explain these deviations. Then, we continue in 5.3 with alternative explanations for the underlying mechanisms behind this deviations. To clarify this section structure, we begin section 5.2 with the following paragraph:

> Ultimately, we want to learn how people respond to the expected *costs* from signals generated by different information structures and whether the simple theoretical model adequately captures this behavior. We therefore first examine whether and how subjects deviate from the benchmark WTP of our risk-neutral, Bayesian-updating, utility-maximizing agent. We then test whether these deviations can be explained by risk preference or updating accuracy. Finally, we describe our findings on the heterogeneous sensitivity of the WTP by prior — which motivates our discussion on the potential mechanisms behind these deviations that we address in the next section.

Hopefully, this reorganization and the addition of the "road map" paragraph sufficiently clarified the link between Result 2 and the subsequent empirical analysis.

b) Similarly, given the importance of Figure 5, it would be nice if the authors could include confidence interval of the regression coefficients, and present in more details the regression specification.

Response: We have now added the confidence intervals onto Figure 5 and details of the regression specification in the main text in Section 5.3.

c) Figure 5 shows the sensitivity to FP and FN are similar and unchanged as prior changes. Does that mean fixing FP and FN, WTP is unchanged in prior? That would violate the intuition that the value of information decreases as the prior becomes more extreme. As I suggest the authors increase the spotlight on Figure 5, it would be a good opportunity to present some analysis on the reported WTP instead of the difference between the reported WTP and the risk-neutral benchmark.

Response: We thank the referee for this valuable suggestion that have improved the clarity of our exposition. Figure 5 shows regression coefficients of WTP with respect to FP and FN rates and does not inform on actual levels of WTP which indeed change with priors. We extended the explanation of Figure 5 in the following way:

> In Figure 5, we overlay these estimates on the line plots of the theoretical sensitivities of WTP on FP and FN rates across different priors (based off Equation 8).[2] The figure shows that the point estimates of the sensitivity

[2]Given that theoretical sensitivities depend on information structures which are pooled within priors, we fit the same regression to the theoretical values and report coefficients as theoretical sensitivities. With tobit estimates, the empirical coefficients are generally higher but are also very close to each other and follow the same pattern. However, we cannot fit them easily for the theoretical values.

> coefficients of WTP on FP rates for lower priors ($p \in \{0.1, 0.2\}$) are below their theoretical benchmarks. In contrast, the point estimates of WTP sensitivity on FN rates for higher priors ($p \in \{0.3, 0.5\}$) are significantly below. Surprisingly, across all priors, the empirical estimates of the sensitivities on FP and FN rates are very close to each other.

Figure 5 shows regression coefficients of WTP with respect to FP and FN rates and does not inform on actual levels of WTP which indeed change with priors.

We also added a short description of how WTP changes with priors on p. 18:

> Willingness-to-pay for signals increases with priors starting from 86 cents average when $\pi = 0.1$ and reaching 2.1 USD for the highest prior of 0.5. This contrasts with the theoretical value that is maximized for priors of 0.3 (the value for $\pi = 0.2$ is just slightly lower). Hence subjects tend to overpay for signals with low priors and underpay for medium priors.

3. Another potential explanation of the results is probability weighting. If somehow subjects exhibit probability weighting to the extent that probability compresses towards 0.25 for all priors, then sensitivity to FP and FN would be equal and is invariant in prior. I don't think this explanation is realistic, but some discussions (to rule it out) would be nice.

   Response: This is an interesting hypothesis and we now added a small subsection to analyze it on p. 24. We find that probability weighting with rank-dependent utility function should make subjects relatively more sensitive to FN rates in WTP tasks and can partially explain our results. At this moment we do not have enough variation in our experiment to exclude this explanation and hence we specifically refer to it in our conclusion and introduction. However, given the evidence of abnormal sensitivities to FP and FN rates in cases in which they should not affect posterior probabilities of adverse events, we still believe that our original explanation remains the most likely one.

4. I find it unclear how subjects are allocated to different treatments. The authors write in the 2nd paragraph page 10: " Subjects go through two different priors and three types of signals". But on Table 1 page 10, there are total 4 types of priors and 7 types of signals. Are subjects randomly allocated to 2 out of the 4 types of prior and 3 out of the 7 types of signals? But the authors write in the 2nd paragraph, page 10: "Subjects go consecutively over all three signal types starting from the honest one for each prior.". That means one of the three signals is from the honest-only composition (3-0-0) for all subjects, and subjects are not randomly assigned to 3 out of the 7 types of signals? What are the sample sizes for each treatments?

   Response: Thank you for this question that help us improve our description of the treatment assignment. We have now added a subsection "Treatments and Treatment Assignment" in Section 3, which describes the assignment process:

   > Subjects were assigned to treatments in the following manner. Each subject was randomly assigned six treatments that combined a pair of priors of either (0.1, 0.3) or (0.2, 0.5) with three information structures. The presentation of the treatments were organized by prior, and the order of the priors — i.e., whether the first was larger than the second — was randomly assigned. For each prior, the first treatment was always the one with a fully truthful information

structure. The order of priors and signals stays constant for each subject across tasks, but can vary between subjects.

5. I am also confused about Table 2 and 3, especially, column 4, the posterior. It seems that the authors have pooled the different 4*7 treatments into 8 different groups, based on the presence of FP and FN, and the hints received by the subjects. The authors write on page 12 that "Column 4 shows the posterior probability of a black ball averaged across all the treatment within a group" How are the posteriors calculated? Is it based on the empirical data, or is it some weighted average of Bayesian posteriors? Similarly, are the share optimal calculated based on the empirical data or the Bayesian posteriors?

   Response: Thank you for this suggestion which prompted us to clarify our writing. We extended the explanation of Column 4 in the paper in the following way: "Column 4 shows the posterior probability of a black ball averaged across all the treatments within a group. We calculate it from priors and FP and FN rates using Bayes formula for each treatment and each signal value and then average across subjects. " We also made sure that we do no use the word "posterior" when referring to subjects' beliefs when explaining Table 4 and extended the note under Table 4 to explain the meaning of the posterior.

6. The authors write: "subjects tend to overvalue false-negative costs for low probability events and overvalue false-positive costs for high probability events." Where do we see that in Table 5? The coefficients of FP costs, FN costs on column 4 and 5 are all positive. Should it be "subjects tend to overvalue more false-positive costs (coeff:0.800 vs 0.204) for low probability events and overvalue more false-negative (coeff: 0.407 vs 0.150) costs for high probability events."? When comparing coefficients, the authors should also report results in statistical tests.

   Response: Thank you for pointing this out! On the second read, we also found this explanation a bit unclear and revised it to the following:

   > As costs increase, subjects should downward-adjust their WTP accordingly. These positive coefficients show that relative to the risk-neutral benchmark, our subjects did not react enough to rising costs (by adjusting their WTPs), resulting in a WTP discrepancy that increases with costs.

   Table 5 now also reports statistical tests for equality of coefficients on FP and FN costs.

7. The authors cited two laboratory experimental papers on the demand for information. I believe there are many more. I do not think the missing literature diminishes the contribution of this paper, but citing them would highlight the novelty of the paper's setup and better position the paper in the existing literature. Here are some examples:

   (a) Intrinsic Information Preferences and Skewness by Yusufcan Masatlioglu, Yeşim Orhun, and Collin Raymond

   (b) How Do People Choose Between Biased Information Sources? Evidence from a Laboratory Experiment by Gary Charness, Ryan Oprea, Sevgi Yuksel

   (c) Beyond Value: on the Role of Symmetry in Demand for Information by Aniol Llorente-Saguer, Santiago Oliveros, Ro'i Zultan

Response: Thank you for highlighting these relevant studies. We have now added a paragraph to describe existing literature on preferences for non-instrumental information on p. 25 in the paper:

> **Non-instrumental information.** There is evidence in the literature of people valuing "non-instrumental information" that does not affect their decisions. For example, Eliaz and Schotter (2010) find that subjects are willing to pay to know the probability of their choice being correct even if this information cannot affect their choice, while Ganguly and Tasoff (2017) document that most people are willing to pay a small amount to know their pre-determined experimental payoffs at the beginning of the experiment rather than at the end. Similarly, Masatlioglu, Orhun and Raymond (2023) reveal a strong and systematic preference for positively skewed information, even when such signals are less informative, suggesting that anticipatory emotions such as hope and anxiety shape information demand independently of instrumental value.[3]

We also cite Charness et al (2021) paper in a different context.

8. I think one of the low-hanging fruits to improve the paper's contribution is to provide more discussion on policy implications related to warning signals. At the moment, the authors only discuss the papers' contribution in the introduction by saying that "currently there is no guidance on what this threshold might be beyond assuming that decision-makers weigh false-positive and false-negative costs equally." Throughout the rest of paper, the authors did not discuss any implications/guidance on the design of warning signals systems.

Response: We extended the discussion of practical implications in the conclusion in the following way:

> Our findings have direct applications for the design of warning signals. Bias in preferences for signals arising from risk aversion means that prioritizing false-negative outcomes while designing warning signals can be welfare enhancing, even when it increases the expected costs of using the signal. At the same time, finding ways to better present the signal's information structure to help users discriminate between false-positive and false-negative payoffs should improve individual decision making. Studies on Bayesian updating, for instance, show that medical professionals make better decisions if information on medical tests is presented in the form of expected frequencies rather than as a tuple of prior conditional probabilities (Hoffrage et al., 2015; McDowell and Jacobs, 2017).
>
> Understanding the source of this asymmetry can also help reduce externalities associated with signal (mis)use. We find that people overvalue signals with excessive false alarms for a typical case with low-probability events. In some cases, individuals may not incur the full cost of false positive signals. For example, the cost of medical overtreatment from tests with high false positives

[3] Unlike Masatlioglu, Orhun and Raymond (2023), Oliveros, Zultan and Llorente-Saguer (2025) find that decision makers exhibit a striking and persistent preference for symmetric sources, i.e., those offering a balanced distribution of positive and negative signals, even when asymmetric sources would yield more decision-relevant information. The authors argue that symmetry exerts an intrinsic appeal, suggesting that individuals prefer information structures that "feel fair" or representative, independent of their actual informativeness.

may be an externality absorbed by the healthcare system. Similarly, first responders may be required by law to respond to automatic fire alarms installed in commercial buildings resulting in excess costs borne by taxpayers when false positive rates are high.

Future research should also explore the role of warning signals as an experience good, to wit, how prolonged exposure to a signal with a given information structure can influence individual biases. For example, buying a smoke alarm that never misses may seem like a good idea at first, but may seem less so after a year experiencing repeated false alarms. Given our finding of insufficient attention to false-positive costs for rare events, ex-ante preferred warning signals could later be considered overly sensitive and ignored, leading to *alarm fatigue* and increasing the risk of potentially preventable disasters.

9. The use of multiple gremlins frames both FP and FN as "dishonest". Thus, one would imagine that subjects refrain from trusting/paying "dishonest" gremlins, leading to equal sensitivity to FP and FN. The authors could consider running new sessions that frame signals as generated from one imperfect gremlin/expert that has asymmetric tendencies to send false positive and false negative signals.

Response: Thank you for this suggestion. Our formulation follows the original framing of Coutts (2019). We agree that this framing could partly explain the results. We therefore carefully considered the trade-off from reframing the description of the gremlins for our new (post R&R) data collection. We believe such a reframing of a key component of our experiment (i.e., how we describe the signals) would change the experiment enough that we would be unable to pool our new data with the original data. This will not only lead to the loss of statistical power, but also limit our ability to provide robust evidence of our results by combining our original treatments with new treatments that provide intermediate values of posteriors and WTPs. We have therefore decided to keep our original framing for the additional data.

10a. There are no "stars" in Table 5 onwards. Is that intentional? I find it impossible to calculate the p-values for every coefficient.

Response: Thank you for pointing this out. We apologize for the absence of the statistical significance stars, which we forgot to put back after being removed due to an editorial requirement of another journal. We have now put the significance stars in all regression tables.

10b. In various parts of the paper, the authors have used the terms "true posterior" and "accurate posteriors" (for example Table 2 and 3, the bottom of page 14, and many others.). Do they refer to Bayesian posterior or empirical posterior?

Response: In both cases we use posterior to refer to Bayesian posteriors calculated from priors and information structures. We no longer use the term "accurate posterior" and added a footnote next to "true posteriors" encountered on p. 14: *Posteriors calculated from priors and information structures using the Bayes formula.*

10c. Page 18 after the regression equation: how exactly are FP cost and FN cost calculated? It is confusing to see FP and FN sometimes refer to rate and sometimes refer to cost.

Response: Thank you for pointing this lack of clarity in our exposition. We have now clarified this when describing our regression equation in Section 5.2:

> We begin by estimating the relationship between the deviations of the empirical WTP from the benchmark model on FP and FN costs. The FP and FN costs studied here — defined above as $\pi(1 - P_{01})c$ and $\pi P_{01} L$ — differ from their respective rates because they incorporate the cost associated with each error type. For example, a high false-negative rate imposes fewer costs when priors are low because the adverse outcome is already unlikely, whereas a high false-negative rate carries substantial costs when the prior probability of the adverse outcome is high. We estimate the following regression:
>
> $$\Delta b_{is} = \beta_0 + \beta_1 FPC + \beta_2 FNC + \delta_i + \varepsilon_{is}$$
>
> where $\Delta b_{is} = (b_{is} - b_s^*)$ is the difference between individual i's WTP and the risk-neutral WTP for signal s, and FPC (FNC) is the false positive (false negative) cost.

11a. Extra "=" sign in Equation (1) page 5.

11b. Missing a fullstop at the end of footnote 7 page 15.

Response: Thank you for pointing out these typos (#11a and b). We have addressed them in the paper.

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